

Safety

The biosafety of our *Escherichia coli* Nissle 1917 (EcN)-based therapeutic system for small-cell lung cancer (SCLC) is built on a **four-layer, closed-loop containment strategy**, adhering to iGEM's safety guidelines and clinical translation principles. This design strictly restricts engineered bacteria to the SCLC tumor microenvironment (TME), eliminates off-target risks, and prevents systemic dissemination or environmental escape.

- Chassis Strain Intrinsic Safety: EcN, a clinically validated probiotic, forms the first safety barrier: it lacks pathogenic virulence factors, has low immunogenicity (avoiding excessive pulmonary inflammation), and is sensitive to common antibiotics (e.g., ciprofloxacin). Its innate tumor tropism and facultative anaerobism enable adaptation to the hypoxic SCLC TME but prevent survival in aerobic normal lung tissue or healthy organs.
- Suicide Switch Design: A dual-promoter *thyA*-dependent suicide switch ensures tumor-specific survival. In vitro, the arabinose-inducible ParaC promoter drives *thyA* expression, supporting robust proliferation. In vivo, expression switches to the lactate-responsive PlldR promoter—only activated in the SCLC TME. In normal tissues with low lactate, *thyA* is silenced, triggering bacterial apoptosis due to thymidine deficiency.
- After the engineered bacteria enter the lungs via nebulization, the activation of their functions is subject to multiple layers of regulation.
 - First, the downstream therapeutic pathway is controlled by a quorum sensing (QS) module, which is only initiated when the bacterial density reaches a predefined threshold, thus preventing random expression at low cell densities.
 - Second, the expression of therapeutic short peptides is dependent on a lactate-responsive module, which is significantly activated exclusively in the high-lactate metabolic microenvironment characteristic of small-cell lung cancer (SCLC). This confines engineered bacteria to exert their effects primarily in tumor regions and minimizes unintended impacts on normal lung tissue.
 - Collectively, these design features ensure that the engineered bacteria achieve only transient, localized, and controllable functional colonization within the lungs.